

mTORC1 activity as a determinant of cancer risk--rationalizing the cancer-preventive effects of adiponectin, metformin, rapamycin, and low-protein ...



Increased plasma levels of adiponectin, metformin therapy of diabetes, rapamycin administration in transplant patients, and lifelong consumption of low-protein plant-based diets have all been linked to decreased risk for various cancers. These benefits may be mediated, at least in part, by down-regulated activity of the mTORC1 complex, a key regulator of protein translation. By boosting the effective availability of the translation initiator eIF4E, mTORC1 activity promotes the translation of a number of "weak" mRNAs that code for proteins, often up-regulated in cancer, that promote cellular proliferation, invasiveness, and angiogenesis, and that abet cancer promotion and chemoresistance by opposing apoptosis. Measures which inhibit eIF4E activity, either directly or indirectly, may have utility not only for cancer prevention, but also for the treatment of many cancers in which eIF4E drives malignancy. Since eIF4E is overexpressed in many cancers, strategies which target eIF4E directly--some of which are now being assessed clinically--may have the broadest efficacy in this regard. Many of the "weak" mRNAs coding for proteins that promote malignant behavior or chemoresistance are regulated transcriptionally by NF-kappaB and/or Stat3, which are active in a high proportion of cancers; thus, regimens concurrently targeting eIF4E, NF-kappaB, and Stat3 may suppress these proteins at both the transcriptional and translational levels, potentially achieving a very marked reduction in their expression. Copyright © 2011 Elsevier Ltd. All rights reserved.